

Manuscript

Layer-1

稿件草稿 HTML 版（基于 docs/manuscript_draft.md）

Self-contained academic HTML with embedded SciencePlots figures (base64).

Literature-anchored Layer-1 surrogate for exploratory screening of indirect mitral annuloplasty

Working short title: FMR IMA Layer-1 exploratory surrogate (software / methods)

Venue class: computational biomechanics / research-software methods paper (CMBBE / MedEngPhys / RSOS-class) — not a clinical decision trial.

One-sentence argument: We provide a transparent, literature-anchored algebraic surrogate that reproduces selected Galili peak-systole anchors, supports Dryad-aware provenance and leave-one-case-out scoring, and ranks exploratory IMA-CS / IMA-AP scenarios under explicit geometric, risk-screening, and η assumption-prior uncertainty — without claiming new LHHM FEA, clinical recommendation, or independent external validation of blended cases.

Data / reproduce: `python run_pipeline.py --seed 42 --paper · python tools/import_galili_dryad.py --from-fixture --process · python tools/loo_evaluate.py --write`

Verification levels: Level 0 = calibrated to / reproduced selected Galili cases; Level 1 = exploratory planning map + uncertainty-aware scenario ranker. Level 2 patient-specific LHHM/FSI is out of scope.

Abstract

Background. Computational IMA studies (Galili et al., *R. Soc. Open Sci.* 2022) report suture/bridge shortening with peak-systolic geometry and leakage. Peak-systolic IMA-AP 50% yields AP **15.9 mm** (near-direct AP effect), not undeformed diastole **34.4 mm**. Device classes differ: MAVERIC evaluates **ARTO** (CS–IAS / IMA-AP-class); Carillon / REDUCE-FMR are IMA-CS-class.

Methods. We implement a reproducible Python Layer-1 surrogate: phenomenological mechanics plus a literature-calibrated leakage proxy. Provenance prefers Dryad doi:10.5061/dryad.bzkh1899d when local archives are present, with published table scalars as a documented secondary path. Calibration vs held-out splits and leave-one-case-out scoring are explicit. A discrete scenario ranker

minimizes physics leakage-proxy regurgitation subject to an AP ceiling, an illustrative NiTi engineering screen, and a Rottländer CS–LCx risk-screening threshold, reporting P(feasible), η ranking stability, and a Pareto frontier (leakage vs AP reduction vs LCx vs strain). Dual-suture commissural $\times 0.5$ is an exploratory hypothesis parameter. Prefer patient-measured CS–LCx; the cinch slope is a labeled assumption.

Results (seed=42). Under the planning map the ranker evaluates 36 points and retains 30 feasible designs ($P(\text{feasible}) \approx 0.86$ on the device grid). Best candidate under stated assumptions: **IMA-AP dual suture 60%** (assumption $\eta_{\text{ap}}=0.30 \rightarrow$ AP reduction **18.0%**, physics leakage-proxy **$\sim 0.074\%$** , jet=central). Best feasible IMA-CS: bridge **20%** (CS–LCx **8.6 mm** at the risk-screen edge). η assumption-prior sensitivity shifts the top-ranked setting (top-1 stability ~ 0.33 under $\pm 20\%$ η sampling); this is not FEA UQ. Held-out / LOO metrics score blend-off predictions against published peak-systole quantities without advertising calibration-blend cases as validated.

Conclusions. A literature-anchored low-order surrogate can support **exploratory screening** of IMA strategy settings with inspectable assumptions and uncertainty language. It is not a preoperative clinical decision system and must not equate physics % with clinical regurgitant volume.

Keywords: functional mitral regurgitation; indirect mitral annuloplasty; surrogate model; exploratory screening; research software; Layer-1

1. Introduction

Functional mitral regurgitation (FMR) reflects ventricular remodeling, annular dilatation, and impaired leaflet coaptation. Indirect mitral annuloplasty (IMA) concepts remodel annular geometry without direct leaflet repair. **IMA-CS** (Carillon-class) and **IMA-AP** (CS–IAS; ARTO / MAVERIC mechanism class) are not mechanically interchangeable.

Galili et al. (2022) compared generic IMA-CS and IMA-AP in the Living Heart Human Model. IMA-AP acts nearly directly on peak-systolic AP (disease 26.1 mm $\rightarrow$ IMA-AP 50% $\rightarrow$ **15.9 mm**). Undeformed diastolic AP 34.4 mm is a different cardiac phase.

What remains for software methods is transparent **dose translation, provenance, and exploratory ranking** under stated assumptions—not a claim of first CS-vs-AP comparison, not new FEA, and not a clinical recommendation engine.

2. Related work and positioning

Galili RSOS 2022 + Dryad supporting files provide the computational leakage/ROA/AP table we anchor to. Clinical series (ARTO/MAVERIC; Carillon TITAN II; REDUCE-FMR) supply **directional / geometric context only**. This paper contributes research-software packaging and an inspectable Layer-1 ranker, not a new high-fidelity FSI study.

3. Software architecture

```
configs/    surrogate_calibration.yaml, design_space.yaml, case YAML
models/     geometry, pathology, devices (Galili peak-systole + planning map)
simulation/ phenomenological mechanics proxy (legacy alias run_fea_surrogate)
sph/        literature-calibrated leakage proxy (SPH-inspired; not a full SPH solver)
analysis/   ROA, jet, sweep, scenario_ranker, paper tables/plots
tools/      import_galili_dryad.py, loo_evaluate.py, package_reports.py
data/       raw drop zone, fixtures, processed galili_cases.csv
```

Runnable entry points: `run_pipeline.py`, `python -m analysis.planner`, `python -m analysis.design_sweep`.

4. Methods

Pathology and geometry. Parametric papillary pathology (44% posterior passive). Undeformed diastole and peak systole are separate cardiac_phase records.

Device kinematics. Galili mode interpolates published peak-systolic AP; clinical/planning mode uses assumption-prior η or preferred ΔAP . η_{CS} is **not** fitted from MAVERIC/ARTO.

Mechanics and leakage proxies. Algebraic coaptation-gap / strain / contact proxy scores. Leakage proxy is literature-calibrated at pathology. Physics regurg % $\neq$ clinical regurgitant volume. Synthetic contact-cluster ROA is visualization-only.

Provenance and LOO. `tools/import_galili_dryad.py` downloads or documents manual drop; extracts independent AP, contact, and leakage descriptors. `tools/loo_evaluate.py` scores held-out and leave-one-case-out predictions with blend OFF.

Scenario ranker. Minimizes physics leakage proxy; reports $P(\text{feasible})$, $\eta \pm 20\%$ ranking stability, and Pareto frontier language. Dual-suture factor is an explicit hypothesis. LCx prefers patient-measured baseline.

5. Results

5.1 Galili peak-systole anchors (reproduction)

Case	Leakage %	Peak-sys AP mm	ROA mm ²
pathology	5.26	26.1	172.8
ima_cs_22	0.29	24.8	48.2
ima_ap_50	0.08	15.9	27.3
ima_ap_70	0.13	12.4	46.1

5.2 Exploratory scenario ranking (planning map, seed=42)

- Evaluated **36** / feasible **30** ($P(\text{feasible}) \approx 0.86$ on the device grid)
- **Best candidate under assumptions:** IMA-AP dual **60%**, AP **18.0%**, physics **~0.074%**, jet=central
- Best IMA-CS: **20%** bridge, CS-LCx **8.6 mm** at risk-screen edge
- Dual-suture jet rule: exploratory hypothesis only
- η uncertainty: ranking stability reported in scenario_ranking.json $\rightarrow$ uncertainty

5.3 Held-out / LOO (blend OFF)

See results/output/loo_evaluation.json. Calibration IDs with blend ON are reproduction only.

5.4 Directionality-only clinical context

ARTO/MAVERIC: AP↓. Carillon TITAN II: ~15% AP context. REDUCE-FMR: regurg↓. Magnitudes not equated to physics %.

6. Discussion / limitations

Central advance: transparent exploratory screening software with corrected Galili peak-systole AP semantics, MAVERIC=ARTO mapping, Dryad-aware provenance, and uncertainty-aware ranking language. Limitations: algebraic proxies; Dryad CDN may require manual zip drop; LOO is not patient-level external validation; η are priors; LCx/NiTi screens are literature/engineering screens not safety clearances.

7. Availability

- Code: <https://github.com/Coucou2016/fmr-ima-layer1-planner>
- License: MIT (LICENSE)
- Citation: CITATION.cff · references.bib
- Model card: MODEL_CARD.md · Data: DATA_PROVENANCE.md

8. Conclusion

A Layer-1 literature-anchored surrogate can rank exploratory IMA scenarios under explicit assumptions after correcting peak-systole Galili AP facts and MAVERIC device-class attribution. Cite as methods/software for exploratory screening—not as clinical preoperative decision support.

Collab *notes:* docs/chatgpt_collab/20260913_p0_review_response.md,
docs/chatgpt_collab/20260914_plp2_implementation.md

Embedded SciencePlots figures

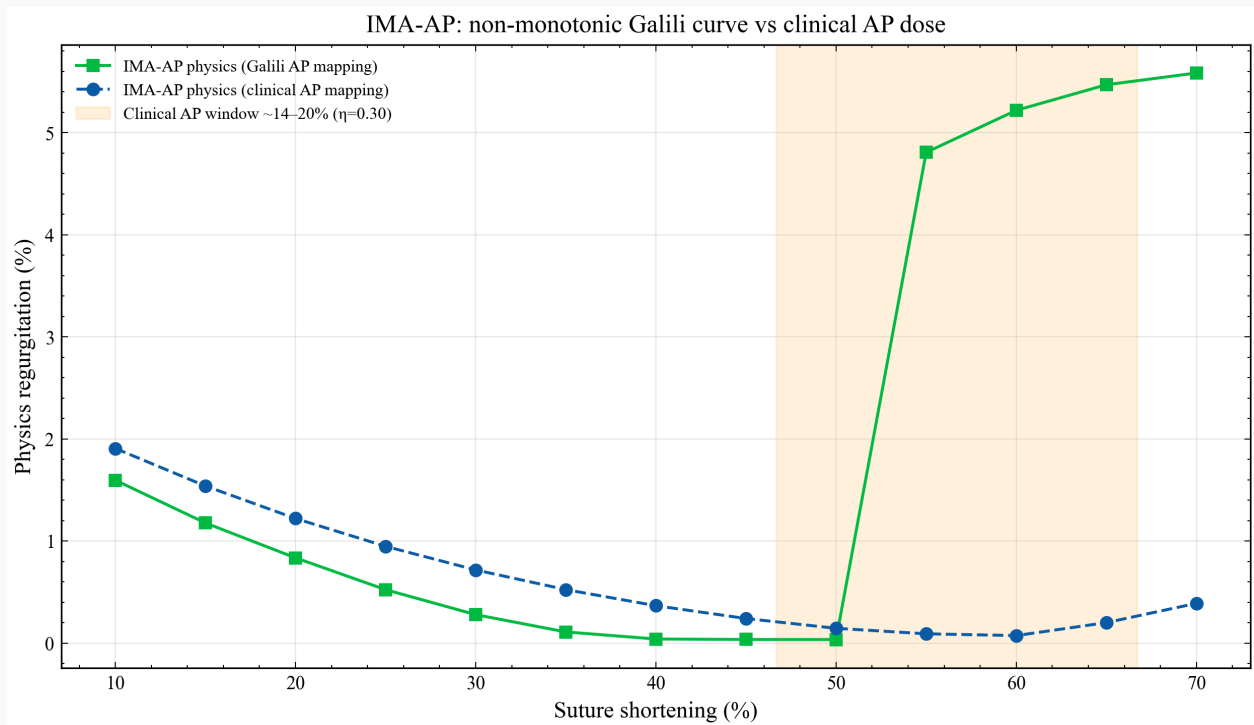


图 1. IMA-AP 物理反流随缝线缩短百分比的变化：Galili 映射 vs 临床映射

文件：fig1_ima_ap_nonmonotonic_clinical_window.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：本图回答“计算缝线缩短百分比能否直接当作临床前后径（AP）剂量？”

横轴为 IMA-AP 缝线缩短 %；

纵轴为 Layer-1 泄漏代理 physics_regurgitation_pct（代数力学代理 + 文献校准泄漏代理；不是超声反流容积分数）。

如何读：（1）Galili 映射复现发表峰缩期 AP：IMA-AP 50% → 峰缩期 AP=15.9 mm（相对疾病 26.1 mm 约 39% AP 缩减），

并非未变形舒张期 34.4 mm / 0% AP；70% → 12.4 mm 并交界区泄漏升高。（2）规划映射采用假设先验

$AP_reduction\% = \eta \times shortening\%$ ，默认 $\eta_{ap}=0.30$ （假设，非临床标定），使 50% 缝线对应约 15% AP，落入

ARTO/MAVERIC（IMA-AP 类，不是 Carillon）约 14–15% 对话窗口。绿色带为约 14–20% AP 窗口。

结论（仅限本代理）：规划映射下窗口内总体可筛查；Galili 峰缩期映射下 70% 非单调恶化。

勿将纵轴等同临床试验反流容积。

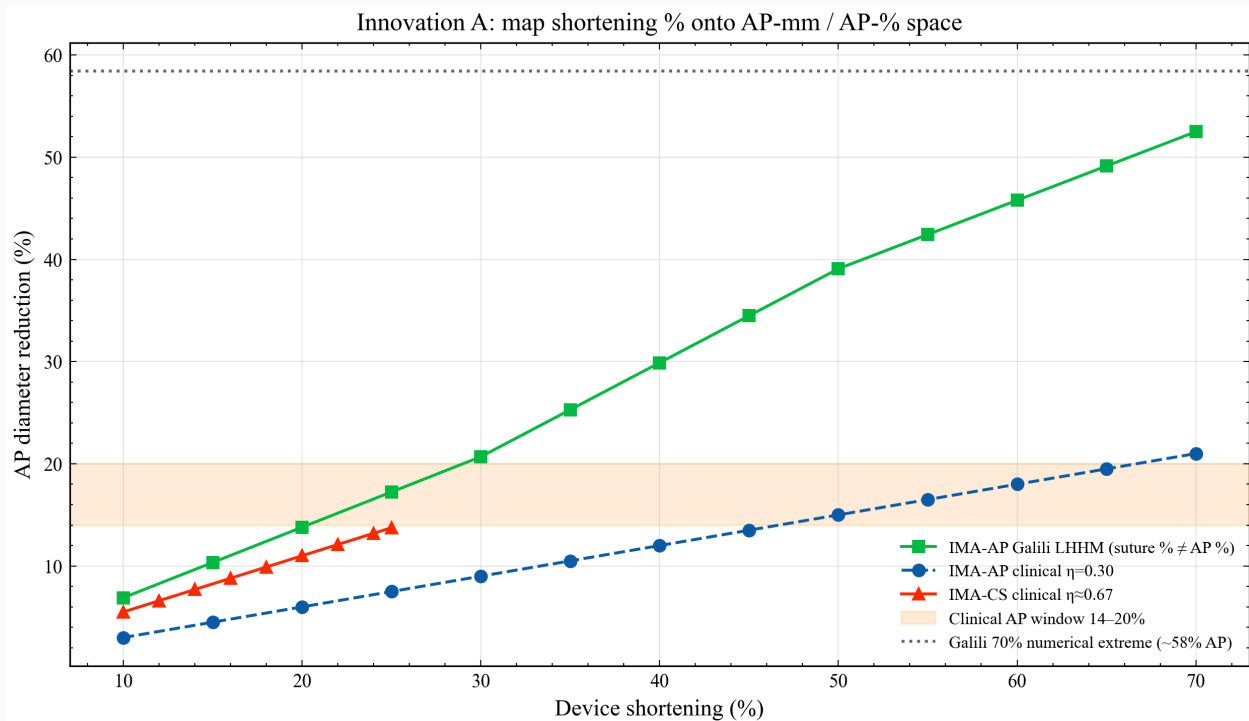


图 2. 装置缩短百分比到 AP 直径缩减百分比的剂量映射

文件: fig2_suture_vs_ap_reduction.png (SciencePlots + Times New Roman, dpi $\geq$ 300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 横轴为缩短 %; 纵轴为 AP 缩减 %。轨迹对照: Galili 峰缩期 (IMA-AP 近直接缩 AP)、

规划 IMA-AP (假设 $\eta_{ap}=0.30$)、规划 IMA-CS (假设 $\eta_{cs}=0.55$, 非 MAVERIC 拟合; Carillon/TITAN II ~15% 仅语境)。

MAVERIC=ARTO (IMA-AP 类)。水平带 14–20% 为探索性规划窗口。

如何读: 同一缩短 % 在不同映射下 AP 缩减截然不同。 η 为假设先验, 不是成像–力学辨识, 也不是“临床标定常数”。

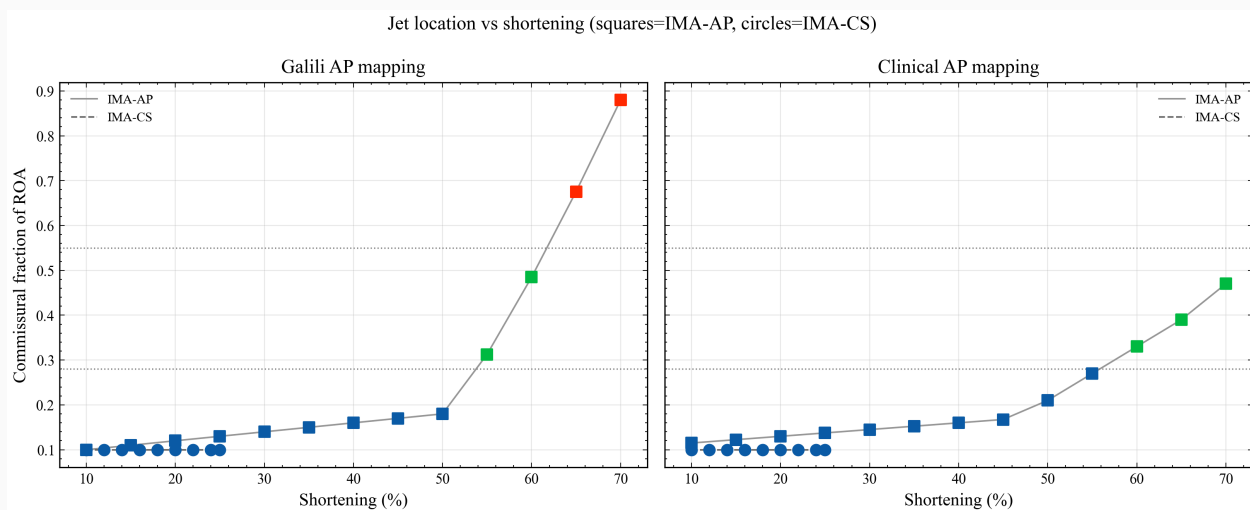


图 3. 射流位置与交界区 ROA 分数随缩短变化（Galili vs 临床）

文件: fig3_jet_location.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 反流“量”之外, 筛查还关心泄漏机制位置。本代理将 ROA (regurgitant orifice area,

反流口面积, 单位 mm^2 , 由对合间隙等代理量估计) 拆为中央份额与交界区份额, 并分类为 central /

commissural / mixed。横轴为缩短 %; 纵轴为交界区 ROA 分数; 方块=IMA-AP, 圆点=IMA-CS; 左右面板对照

Galili 与临床映射。水平虚线为分类阈值示意。双缝线 $\times 0.5$ 为显式假设参数, 不是独立“发现”。

如何读: 交界区分数升高意味着泄漏更偏交界区; 分类从 central 变为 mixed/commissural 表示机制标签切换。

该标签是代理口上的机制草图, 不是超声 PISA (proximal isovelocity surface area) 或多普勒诊断。

seed-42 假设下最优候选双缝线 60% 的机制来源可在本图与图 5 对照中追溯: 同 AP 下交界份额更低、jet 保持 central。

结论: 在临床映射下, IMA-AP 高剂量单缝线更易出现 mixed; IMA-CS 在可行桥缩短范围内多保持 central。

结合图 5, 双缝线可在匹配 AP 缩减下压低交界区分数 (假设情景)。

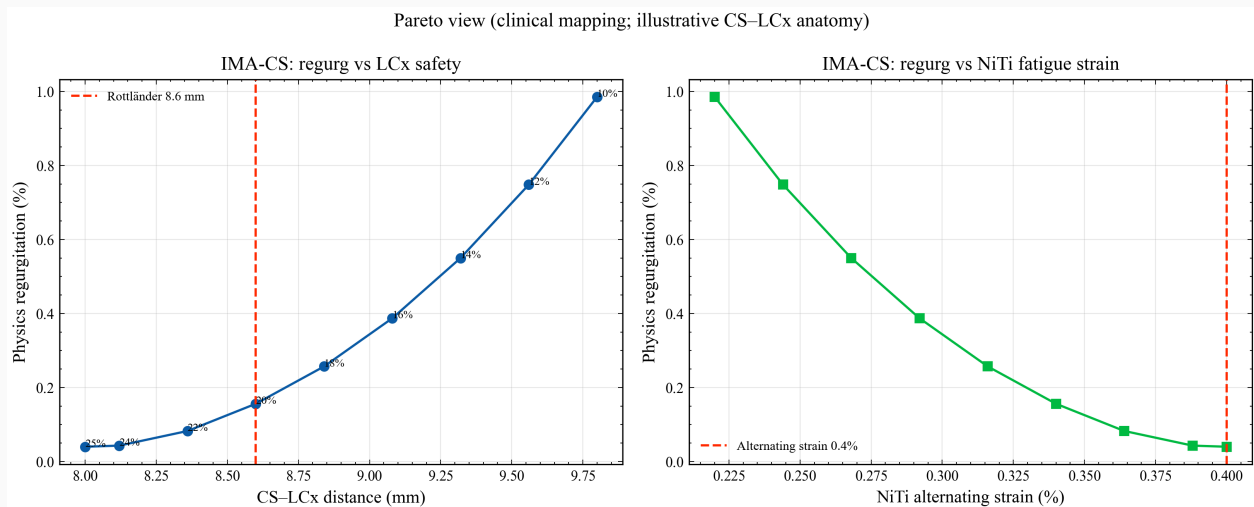


图 4. IMA-CS 临床映射：反流-筛查 Pareto（CS-LCx 与 NiTi 交变应变）

文件：fig4_pareto_lcx_strain.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：IMA-CS 在降低反流的同时可能压缩 CS-LCx 距离（coronary sinus-left circumflex artery，

冠状窦与左回旋支间距）。文献筛查边界取远端着陆区 CS-LCx ≥ 8.6 mm（Rottländer 等；注意：<8.6 mm

预测妥协 ≠ ≥8.6 mm 已证明安全）。同时对 NiTi（镍钛合金桥）交变应变设置 <0.4% 的工程筛查上限。

默认示意解剖基线 CS-LCx=11.0 mm，可替换为患者 CT。标题用“筛查”而非“安全证明”。

如何读左右面板：横轴多为筛查指标（CS-LCx 或交变应变），纵轴为 physics 反流；点随桥缩短移动，

越接近约束边界，可行域越窄。22% 及以上在默认解剖上常因 CS-LCx 违约束而不可行。

结论：默认解剖上可行最优贴近 CS-LCx=8.6 mm 的桥缩短 20%。该结果是规划器筛查信号，不是患者级安全证明。

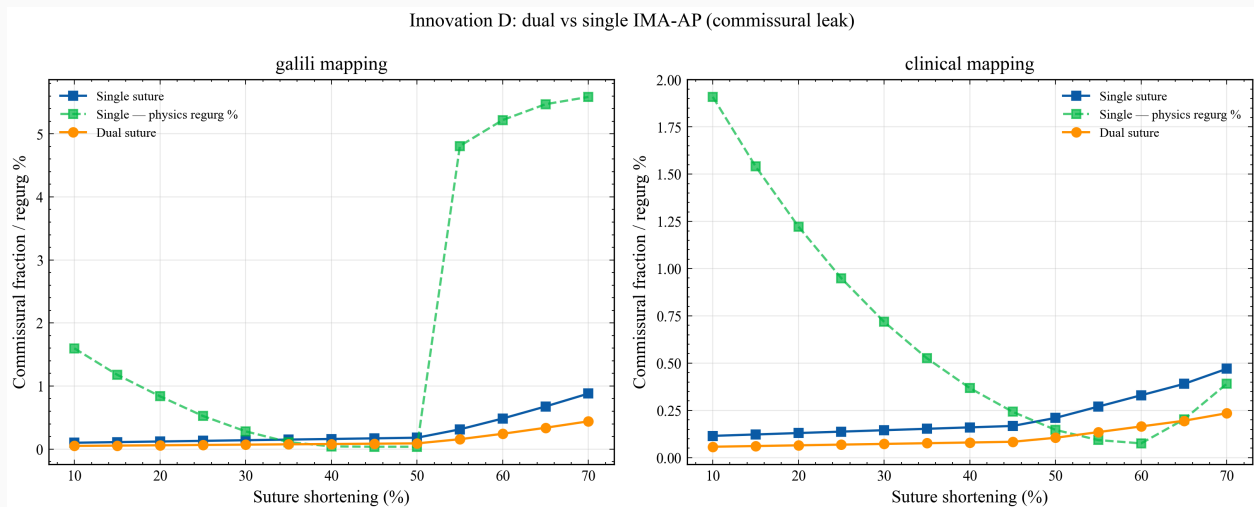


图 5. 双缝线 vs 单缝线 IMA-AP：匹配 AP 缩减下的交界区泄漏对照

文件：fig5_dual_vs_single_suture.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：可选贡献 C4（Innovation D）在相同缝线缩短 %（故相同临床 AP 缩减）下比较单/双缝线。

纵轴关注交界区 ROA 分数；虚线可叠加单缝线 physics 反流作对照。左右面板对照 Galili / 临床映射。

本图隔离“缝线数量”变量，回答机制问题，不是器械清关或产品声明。

如何读：在 60% 临床映射点，单缝线 jet=mixed、交界分数 0.330；双缝线 jet=central、交界分数 0.165，

physics 泄漏代理理由约 0.075% 降至约 0.074%——分类改善更醒目，幅度改善相对温和。70% 点 AP 缩减 21%

超出默认 20% 规划上限，仅作机制对照，不作植入候选声明。双缝线交界因子 $\times 0.5$ 为显式假设参数。

结论：双缝线是假设生成情景；seed-42 场景排序最优候选双缝线 60% 正是在该对照下选出的网格点。

勿将 $\sim 0.074\%$ physics 等同于临床反流容积改善。